

POINT RETREAT, AK, USA

WMO: 997797

Lat: 58.411N

Lon: 134.955W

Elev: 105

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.1	(c) 16.6	(d) 4.4	(e) 6.9	(f) 16.4	(g) 7.6	(h) 8.2	(i) 19.5	(j) 50.3	(k) 16.2	(l) 43.3	(m) 18.0	(n) 29.5	(o) 320	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	69.3	59.8	66.2	58.1	63.4	57.0	61.0	67.2	59.3	64.7	58.0	62.3	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	72.6	63.2	56.8	69.0	61.3	55.7	66.3	59.9	27.0	67.2	26.0	64.7	25.0	62.7	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 35.5	(b) 30.4	(c) 25.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	32.0	32.3	35.9	41.1	48.4	54.1	57.6	57.1	51.6	44.3	37.5	33.6		
	DBStd	10.59	8.06	6.52	5.82	3.76	4.18	3.83	4.12	3.62	3.33	3.24	5.02	6.30		
	HDD50	2927	557	496	438	267	82	5	0	0	18	179	375	510		
	HDD65	7720	1022	916	903	717	514	327	234	245	401	642	825	974		
	CDD50	690	0	0	0	1	33	130	235	221	67	2	0	1		
	CDD65	7	0	0	0	0	0	2	4	1	0	0	0	0		
	CDH74	17	0	0	0	0	0	4	7	6	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	9.3	14.0	13.3	11.7	7.3	6.0	6.0	5.6	5.7	6.9	9.4	12.5	12.9		
(15) Precipitation	PrecAvg	81.1	6.7	5.1	4.7	4.4	4.3	4.0	6.1	7.4	12.0	10.5	8.3	7.8		
	PrecMax	98.5	13.1	8.8	7.0	8.3	9.0	7.8	11.2	12.7	17.2	17.8	16.3	14.3		
	PrecMin	61.3	2.4	1.9	1.7	1.7	1.5	1.1	2.5	2.9	7.0	5.2	3.5	3.4		
	PrecStd	9.6	2.6	2.0	1.4	1.7	1.9	1.6	2.1	2.7	2.4	2.7	2.9	3.0		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.7	44.1	50.2	56.0	65.4	72.6	74.6	72.7	64.1	54.2	48.1	53.4		
		MCWB	44.7	41.1	43.7	48.1	53.7	59.8	62.1	60.9	56.3	52.3	46.0	47.9		
	2%	DB	45.7	42.3	45.7	51.4	61.2	65.9	70.0	68.5	60.7	52.0	46.5	43.6		
		MCWB	42.1	39.8	41.7	44.4	52.4	57.0	60.7	59.0	53.8	49.9	44.6	40.8		
	5%	DB	42.5	40.9	43.3	48.7	57.5	62.6	66.8	65.3	58.0	50.2	45.3	41.9		
		MCWB	40.3	38.7	39.5	43.0	50.5	55.1	59.2	58.0	53.3	48.0	43.5	39.8		
	10%	DB	40.6	39.3	41.8	46.4	54.6	59.8	63.6	62.4	56.2	48.7	43.7	40.8		
		MCWB	38.8	37.3	38.6	41.9	48.8	53.6	58.1	56.9	52.9	46.5	41.8	39.0		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.9	42.0	45.6	48.4	55.0	61.1	64.4	63.3	57.4	52.9	46.6	48.2		
		MCDB	49.4	43.3	49.8	55.1	62.2	70.6	72.1	68.5	61.5	53.5	47.2	53.6		
	2%	WB	42.3	40.4	42.2	45.5	53.1	58.3	61.7	60.4	55.3	50.4	45.3	41.6		
		MCDB	44.7	41.7	44.7	49.9	59.9	64.8	67.7	66.4	57.6	51.5	46.2	42.8		
	5%	WB	40.7	39.1	40.6	43.9	51.2	56.0	59.9	59.1	54.3	48.6	43.9	40.3		
		MCDB	42.5	40.5	42.7	47.6	56.5	60.9	65.3	64.0	56.6	49.9	45.1	41.5		
	10%	WB	39.1	37.8	39.2	42.5	49.5	54.4	58.4	57.7	53.2	47.0	42.1	39.1		
		MCDB	40.6	39.0	41.1	45.7	53.8	58.5	63.0	61.5	55.5	48.5	43.3	40.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.1	6.2	7.9	9.6	9.6	8.7	7.9	6.5	4.6	4.6	4.5		
		MCDBR	8.9	5.8	8.2	11.7	14.5	14.4	14.1	13.0	10.1	6.0	5.0	6.4		
	5% WB	MCWBR	6.9	5.1	5.8	7.1	8.0	8.2	7.8	7.4	6.1	5.2	5.0	5.6		
		MCWBR	8.5	4.9	7.5	10.3	13.0	12.5	12.6	11.7	8.0	5.5	4.7	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.333	0.324	0.383	0.401	0.412	0.416	0.379	0.363	0.329	0.285	0.262		
	taud		2.100	2.086	2.204	2.069	2.142	2.231	2.267	2.407	2.401	2.361	2.155	2.065		
	Ebn at Noon		162	209	258	258	262	261	257	260	245	217	164	132		
	Edh at Noon		16	27	32	43	43	41	38	31	27	21	15	12		
(44) All-Sky Solar Radiation	RadAvg		120	351	737	1229	1571	1571	1341	1096	697	365	163	83		
	RadStd		15	65	100	141	175	116	148	140	115	57	26	10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	+2.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page